

TSINGHUA UNIVERSITY

交叉信息研究院

ALGORITHM ECONOMICS (80470223-0) — COURSE PROJECT

Probing Self-Resolving Prediction Markets: A Simulation and Empirical Study of Reference-Based Resolution

Author:

Vilhjalmur Arnar Vilhjalmsson (威廉)

2025403455

weil25@mails.tsinghua.edu.cn

20th June 2026

Table of Contents

List of Figures	ii
List of Tables	ii
1 Introduction	1
1.1 Background: forecasting things that never resolve	1
1.2 Context: previous approaches	1
1.3 The problem: the gap	2
1.4 The solution: a well-informed reference replaces the outcome	2
1.5 The proof: summary of the evaluation	2
1.6 Contributions	2
2 Related Work	3
3 Preliminaries: Model, Mechanism, and Assumptions	4
3.1 Setting	4
3.2 Scoring rules: from outcome to reference	4
3.3 The self-resolving mechanism	4
3.4 Assumptions	5
3.5 What the paper proves	6
3.6 Problem statement and research questions	6
4 Methods	6
4.1 An agent-based simulator of the mechanism	7
4.2 RQ1: estimating the profit-maximising deviation	7
4.3 Reproducing the analytic bound $k_{\min}$	7
4.4 Supporting synthetic studies	8
4.5 RQ2: a counterfactual on real Polymarket data	8
5 Experiments and Results	9
5.1 RQ1: truthfulness requires an informed reference	9
5.2 Supporting studies: aggregation behaviour	10
5.3 RQ2: do self-resolving payouts track verifiable payouts on real data?	11
6 Conclusion, Discussion, and Limitations	14
6.1 Summary	14
6.2 Discussion	14

6.3	Limitations	14
6.4	Future work	15
Bibliography		16

List of Figures

1	The self-resolving prediction market mechanism	5
2	RQ1: profit-maximising deviation vs. reference substitutes	10
3	Analytic minimum informational substitutes $k_{\min}$	10
4	RQ2: agreement vs. reference lateness	12
5	RQ2: agreement vs. relative time to close	12
6	RQ2: worked example	13

List of Tables

1	Value of near-oracle agents	11
2	Mixed-population behaviour	11
3	Reference sweep on Polymarket data	12
4	Per-market agreement	13

Abstract

Prediction markets aggregate dispersed beliefs into a single forecast, but their incentives normally rely on a verifiable outcome: agents are eventually paid based on what actually happened. Many decisions of interest — the causal effect of a policy, whether a piece of content violates a rule, the long-run consequence of a choice — never resolve cleanly, so this lever is unavailable. Srinivasan et al. (2025) propose a *self-resolving* prediction market that removes the outcome entirely: agents report sequentially, the market stops at a random time, and every agent is paid the cross-entropy of their report against a late *reference* agent rather than against the truth. Their key insight is that a reference with access to enough conditionally independent signals becomes a good proxy for the ground truth, which makes truthful reporting a (strict) perfect Bayesian equilibrium. Their guarantees are, however, entirely theoretical. This paper reproduces the mechanism in an open-source agent-based simulator and stress-tests it along two axes. First, in a clean synthetic world where the paper’s assumptions hold by construction, we estimate a focal agent’s profit-maximising deviation by Monte Carlo and confirm that the best response converges to truthful reporting as the reference’s number of informational substitutes k grows ($\gamma^* \rightarrow 1$ by $k \approx 64$), operationalising the paper’s main theorems; we also reproduce the analytic minimum-substitutes bound $k_{\min}$. Second, we run a counterfactual study on roughly 90 resolved Polymarket markets, treating each price path as a stream of reports and paying it both ways — self-resolving (against a late reference price) and verifiable (against the realised outcome). The two payment schemes agree almost perfectly when the reference is genuinely late (Pearson ≈ 1.00 , payout ratio ≈ 0.95) and degrade gracefully as the reference is moved earlier, isolating the outcome-leakage channel that the reference relies on. Together the results give empirical content to a purely theoretical mechanism and map out where reference-based resolution succeeds and where it breaks.

1 Introduction

1.1 Background: forecasting things that never resolve

Prediction markets are one of the most successful practical tools for aggregating dispersed information. By letting people bet on the outcome of an event and paying them once the truth is revealed, markets such as the Iowa Electronic Markets and, more recently, Polymarket (Polymarket 2025) produce forecasts that are routinely competitive with polls and expert panels (Berg et al. 2008). The engine underneath is a *proper scoring rule*: a payment scheme that makes honestly reporting one’s true belief the reward-maximising action, provided the outcome is eventually observed (Brier 1950; Gneiting and Raftery 2007; Good 1952).

The catch is the phrase “provided the outcome is eventually observed”. Many of the questions decision-makers care about most are *unverifiable*: the ground truth is hard, expensive, or simply impossible to access. What was the causal effect of a policy that was actually enacted, relative to the counterfactual that was not? Is a given piece of online content genuinely misinformation? What will the long-run social consequence of a choice be? In each case people must still pool scattered information and commit to a “best guess”, yet there is no future observation to pay against. The standard prediction-market lever is missing.

1.2 Context: previous approaches

Two research traditions speak to this problem but neither solves it cleanly. The first is *Aumann’s agreement theorem* and its descendants (Aumann 1976; Kong and Schoenebeck 2022): rational Bayesians who share a common prior and repeatedly exchange beliefs must eventually agree, and — when their signals are informational substitutes (Chen and Waggoner 2016) — that consensus aggregates all of their private information. This machinery aggregates beliefs but assumes agents *report honestly*; it provides no *incentive* to reveal a privately held signal. The second tradition is *information elicitation without verification*, which includes peer prediction (Miller et al. 2005), the Bayesian truth serum (Prelec 2004), and output-agreement mechanisms (Waggoner and Chen 2014). These provide incentives without an outcome, but they typically target *elicitation* from a few

agents rather than efficient *aggregation* from a large pool, and many of them admit uninformative equilibria in which everyone reports the same uninformative thing.

1.3 The problem: the gap

There is therefore a gap between these literatures. *However*, the agreement literature aggregates information but does not incentivise truthful revelation. *Despite* the elicitation literature providing incentives without ground truth, it does not aggregate efficiently and is fragile to uninformative equilibria. What is missing is a single mechanism that simultaneously (i) elicits truthful private beliefs, (ii) aggregates them into one consensus forecast, and (iii) requires *no* access to the outcome — all under roughly the same assumptions that make ordinary prediction markets work.

1.4 The solution: a well-informed reference replaces the outcome

Srinivasan et al. (2025) close this gap with a *self-resolving prediction market*. Agents report sequentially; after each report the market terminates with probability α ; and instead of paying agents against the outcome, the mechanism pays them the cross-entropy of their report against the *final (reference) agent*, who — having observed the full market history — is the best-informed participant. The key insight is one of *informational smallness*: if the reference agent has access to enough conditionally independent signals (“informational substitutes”), then no single earlier agent can meaningfully move the reference’s posterior, so the reference behaves like a noisy but unbiased proxy for the ground truth. Paying against this proxy recovers the truthful incentives of a standard market *without ever observing the outcome*.

1.5 The proof: summary of the evaluation

The original paper establishes that truthful reporting is a strict perfect Bayesian equilibrium under a standard set of assumptions, but it offers no empirical evaluation: the mechanism is never simulated and never run on data. This paper supplies that missing evaluation. We build an agent-based simulator in which the paper’s assumptions hold by construction, and we ask two questions. **RQ1 (incentives, simulation)**: under strategic deviation, when is truthful reporting actually a best response, and how does this depend on how well-informed the reference is? We find that a focal agent’s profit-maximising deviation γ^* converges to the truthful value 1 as the reference’s number of informational substitutes k grows — pegged at gross over-reporting for $k \leq 8$, $\gamma^* \approx 1.2$ at $k = 32$, and exactly 1.0 for $k \geq 64$ — operationalising the paper’s Theorems 1–3. **RQ2 (robustness, empirical)**: on roughly 90 resolved Polymarket markets we pay each price path both self-resolvingly (against a late reference price) and verifiably (against the realised outcome), and find that the two schemes agree almost perfectly at a genuinely late reference (Pearson ≈ 1.00 , payout ratio ≈ 0.95) and degrade gracefully as the reference is moved earlier — exactly tracing out the outcome-leakage channel the mechanism depends on.

1.6 Contributions

Concretely, this project contributes:

1. **An open-source agent-based simulator (srpm)** of the self-resolving prediction market, with configurable agents, signal structures, scoring rules, and strategies, in which the paper’s rationality, common-prior, distinct-signal, and conditional-independence assumptions hold by construction (Section 4).
2. **An incentive (best-response) study (RQ1)** that estimates a focal agent’s profit-maximising deviation by Monte Carlo and shows it converges to truthful reporting as the reference’s informational substitutes grow, together with an exact reproduction of the paper’s analytic minimum-substitutes bound $k_{\min}$ (Sections 4–5).

-
3. **Supporting synthetic studies** confirming that under truthful play the market reproduces the full-information Bayesian posterior to machine precision, that a few near-oracle references pin the aggregate, and that the mechanism pays almost nothing in uninformative equilibria (Section 5).
 4. **A counterfactual empirical study (RQ2)** on real resolved Polymarket data that quantifies how closely self-resolving payouts track the verifiable payouts they are meant to replace, and isolates outcome leakage as the mechanism behind the residual gap (Section 5).

The remainder of the paper is organised as follows. Section 2 reviews related work; Section 3 sets up the model, the mechanism, and its assumptions; Section 4 describes our simulator and experimental design; Section 5 presents the results; and Section 6 discusses limitations and concludes.

2 Related Work

The mechanism studied here sits at the intersection of three literatures, following the framing of Srinivasan et al. (2025). We summarise each and place our own contribution — an *empirical* evaluation — relative to them.

Aumann’s agreement theorem and information aggregation. Aumann (1976) showed that two Bayesians with a common prior cannot “agree to disagree”: common knowledge of their posteriors forces those posteriors to coincide. Subsequent work extended this to repeated belief exchange among many agents, who must eventually reach consensus. Chen and Waggoner (2016) and Kong and Schoenebeck (2022) sharpen this further: when agents’ signals are *informational substitutes* — conditionally independent given the outcome, so each additional signal has diminishing marginal value — the eventual consensus does not merely agree but *aggregates* all of the agents’ private information. This line of work tells us *how* honest Bayesians pool information, but it assumes honesty; it does not ask how to *incentivise* agents to reveal their true signal in the first place. The self-resolving market can be read as supplying exactly those incentives on top of the Aumannian protocol.

Prediction markets and (market) scoring rules. When the designer eventually observes the ground truth, truthful elicitation is a solved problem: *proper scoring rules* reward reports that match the realised outcome and make honesty optimal (Brier 1950; Gneiting and Raftery 2007; Good 1952). Hanson (2003) extended scoring rules into *market scoring rules* (MSRs), which let a sequence of agents each update a shared forecast and be paid for how much they *improve* it; Chen and Pennock (2010) relate this cost-function view of automated market makers to the scoring-rule view, and Chen, Dimitrov et al. (2010) analyse the equilibrium strategies such markets induce. All of these mechanisms, however, ultimately pay out against an observed outcome. The self-resolving market keeps the MSR machinery — in fact a cross-entropy MSR — but swaps the outcome for a reference agent’s report, which is the move we test empirically.

Information elicitation without verification (peer prediction). The peer-prediction literature confronts the no-ground-truth case directly. Miller et al. (2005) pay an agent based on how well their report predicts a peer’s report; the Bayesian truth serum of Prelec (2004) scores reports against the population’s predicted distribution; and output-agreement mechanisms (Waggoner and Chen 2014) reward matching a reference agent. Information-theoretic designs (Kong and Schoenebeck 2019) unify many of these and make truth-telling reward-maximising. The recurring difficulties in this literature are (i) *aggregation* — most mechanisms elicit from a handful of agents rather than distilling a large pool into one forecast — and (ii) *uninformative equilibria*, in which everyone reports the same prior-based answer and still gets paid. The self-resolving market is best understood as a peer-prediction mechanism engineered, via informational smallness, to also aggregate efficiently and to pay (close to) zero in uninformative equilibria.

Where this project fits. Srinivasan et al. (2025) prove that, under their assumptions, truthful reporting is a strict perfect Bayesian equilibrium, and that the gain from any deviation vanishes as the reference accumulates informational substitutes. What they do *not* provide is any empirical or simulation evidence: the mechanism is analysed but never instantiated. Empirical evaluation of related elicitation mechanisms has historically been limited and mixed, often hinging on design details such as the presence of uninformative or effortless agents. Our contribution is to fill this gap for the self-resolving market specifically: we reproduce the mechanism in simulation to check *whether the incentive story actually plays out* when an agent is free to deviate, and we run a counterfactual on real prediction-market data to check *whether the reference-based payouts track the outcome-based payouts* they are meant to replace.

3 Preliminaries: Model, Mechanism, and Assumptions

This section sets up the model of Srinivasan et al. (2025) that our experiments are built on. We keep the notation light and refer to the original paper for full proofs.

3.1 Setting

The goal is to elicit and aggregate probabilistic predictions about a binary outcome $Y \in \{0, 1\}$ (categorical and discretised-continuous outcomes reduce to this case). There are M agents who arrive *one at a time*. Agent t has a private signal $X^{(t)}$ drawn from a common prior $\mathbb{P}(Y, X^{(1)}, \dots, X^{(M)})$ and, on arrival, also observes all previous reports $\mathbf{q}^{(1:t-1)}$. Conditioning on $\hat{X}^{(t)} = \{X^{(t)}, \mathbf{q}^{(1:t-1)}\}$, the agent forms a posterior $\bar{\mathbf{p}}^{(t)} = \mathbb{P}(Y \mid \hat{X}^{(t)})$ and reports some $\mathbf{q}^{(t)} \in \Delta\Omega_Y$ to the mechanism. A truthful agent reports $\mathbf{q}^{(t)} = \bar{\mathbf{p}}^{(t)}$; a strategic agent may report something else.

We write $y_1 = \mathbb{P}(Y = 1)$ and $y_0 = 1 - y_1$ for the prior, and use t for the agent under consideration and r for its *reference agent*.

3.2 Scoring rules: from outcome to reference

A proper scoring rule $S(i, \mathbf{q})$ rewards a probabilistic report $\mathbf{q}$ against a realised outcome i , and is *proper* when reporting one’s true belief maximises expected reward (Gneiting and Raftery 2007). The canonical choice is the *log scoring rule* $S(i, \mathbf{q}) = \log q_i$, which underlies the cross-entropy rule used throughout the paper. The crucial move is to score a report against another *prediction* $\mathbf{r} \in \Delta\Omega_Y$ instead of an outcome:

$$S_{CE}(\mathbf{r}, \mathbf{q}) = -H(\mathbf{r}, \mathbf{q}) = \sum_i r_i \log q_i. \quad (1)$$

Because an agent does not know what the reference will report, it maximises its *expected* cross-entropy, which is maximised by reporting its expectation of the reference’s prediction, $\mathbb{E}[\mathbf{r}]$. The mechanism uses the sequential, shared version of this rule — the *cross-entropy market scoring rule (CE-MSR)* — which pays an agent for how much it *improves* the running forecast relative to the reference:

$$S_{CEM}(\mathbf{r}, \mathbf{q}^{(t)}, \mathbf{q}^{(t-1)}) = -H(\mathbf{r}, \mathbf{q}^{(t)}) + H(\mathbf{r}, \mathbf{q}^{(t-1)}) = \sum_i r_i \log \left(\frac{q_i^{(t)}}{q_i^{(t-1)}} \right). \quad (2)$$

Intuitively, (2) asks: *how much closer to the reference is the new report than the previous one?* Swapping the outcome i in $\log q_i$ for a reference $\mathbf{r}$ in (1) is precisely what lets the market resolve without an outcome.

3.3 The self-resolving mechanism

The mechanism (Figure 1) proceeds as follows:

1. Agents $1, 2, \dots$ arrive in sequence; agent t observes the history $\mathbf{q}^{(1:t-1)}$ and its signal $X^{(t)}$, and reports $\mathbf{q}^{(t)}$.
2. After each report the market *terminates with probability* α . Let T be the (random) terminal time. The terminal agent T is the **reference agent** r : having seen the whole history, it is the best-informed participant.
3. Agents $1, \dots, T - k$ are paid the CE-MSR (2) against the reference's report $\mathbf{q}^{(T)}$. The last k agents (those too close to the reference) instead receive a flat fee R , so they cannot be paid to manipulate a reference that they themselves heavily influence.

The single design parameter k controls how far an agent must be from the reference to be scored against it; larger k means a better-informed reference relative to any single scored agent, at the cost of paying more agents the flat fee.

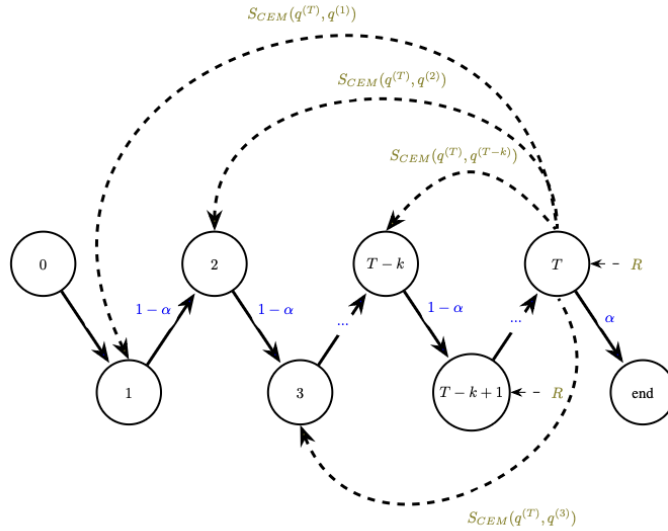


Figure 1: The self-resolving prediction market. Each node is an agent reporting to the mechanism; the market terminates with probability α after each report. The first $T - k$ agents are paid the cross-entropy market scoring rule against the terminal reference agent T , while the last k agents receive a flat fee R . Redrawn after Srinivasan et al. (2025), Figure 1.

3.4 Assumptions

The mechanism's guarantees rest on five assumptions about agents and signals. They matter for us because our two studies are organised precisely around when they hold and when they do not.

- A1 (Rationality & risk-neutrality).** Agents are risk-neutral, fully rational Bayesians, and this is common knowledge.
- A2 (Common prior).** Signals and outcome are drawn from a common prior that is common knowledge.
- A3 (Stochastic relevance / distinct signals).** Each signal induces a *unique* posterior over Y , so agents can invert one another's reports back into the underlying signal.
- A4 (Conditional independence).** Signals are independent given Y — i.e. they are informational *substitutes* (Chen and Waggoner 2016). This is the same class of information struc-

tures under which *standard* prediction markets are incentive-compatible (Chen, Dimitrov et al. 2010).

A5 ((δ, η)-informative signals). Two floors on signal informativeness: δ lower-bounds how much each signal separates $Y = 0$ from $Y = 1$ (via the Bhattacharyya coefficient, $BC \leq 1 - \delta$), and η lower-bounds every signal likelihood, $\min_{x, \omega} \mathbb{P}(X = x \mid Y = \omega) \geq \eta$, so no single report can be arbitrarily decisive.

Two derived quantities recur below. The *Bhattacharyya coefficient* $BC = \sum_i \sqrt{\mathbb{P}(X_i \mid Y=0) \mathbb{P}(X_i \mid Y=1)} = 1 - \delta$ measures the overlap between the two conditional signal distributions ($\delta = 0$ is a useless signal, larger δ a more separating one). The τ -granularity is the smallest gap between the log-likelihood ratios of two distinct signals, which controls how strictly an agent prefers truthful reporting.

3.5 What the paper proves

Three results, all expressing the same idea — *a well-informed reference makes honesty optimal* — frame our experiments:

- **Theorem 1 (bounded influence).** If the reference observes k informational substitutes that agent t does not, then t 's influence on the reference's posterior — the adjustment term Δ — shrinks geometrically, $|\Delta| \leq \frac{1}{4} \left(\frac{1-\eta}{\eta} - \frac{\eta}{1-\eta} \right) (1 - \delta)^k$. Enough substitutes $\Rightarrow$ the reference is effectively ground truth.
- **Theorem 2 (strict truthfulness).** Paid by CE-MSR against a truthful reference, agent t *strictly* maximises expected payoff by reporting truthfully once the reference has at least an explicit threshold number of signals (in δ, η, τ and the prior).
- **Theorem 3 (vanishing gain from deviation).** The gain from *any* deviation is bounded by a function $\hat{D}_\eta(\Delta, \mathbf{y})$ of the adjustment Δ ; since $\Delta \rightarrow 0$ as k grows (Theorem 1), this bound $\rightarrow 0$ — and, crucially, without needing to know τ .

Theorem 1 motivates the analytic minimum-substitutes quantity $k_{\min}$ that we reproduce in Section 4, and Theorem 3 is exactly the prediction our simulation tests: the profit-maximising deviation should collapse onto truthful reporting as k grows.

3.6 Problem statement and research questions

The original paper stops at these guarantees. Our problem is to give them empirical content and to probe their boundaries, through two questions:

- RQ1 (incentives, simulation).** In a world where A1–A4 hold by construction, when does a freely deviating agent actually find truthful reporting to be a best response, and how does this depend on how informed the reference is (the one assumption, A5, that we vary)?
- RQ2 (robustness, empirical).** On real resolved markets — where essentially none of A1–A5 hold — how closely do self-resolving payouts track the standard verifiable (outcome-based) payouts they are designed to replace?

4 Methods

We evaluate the mechanism with two complementary instruments: an agent-based simulator in which the paper's assumptions hold by construction (RQ1 and supporting studies), and a counterfactual replay of real Polymarket price paths (RQ2). Both are implemented in a single Python package, `srpm`, and all figures and tables in Section 5 are produced by the scripts described here.

4.1 An agent-based simulator of the mechanism

The simulator instantiates the model of Section 3 directly. Its components are:

- **Signals.** A `SignalStructure` is a conditional probability table $\mathbb{P}(X \mid Y)$. The two we use are `binary_symmetric( $q$ )`, a signal correct with probability q , and `ground_truth( $\rho$ )`, a near-oracle that reveals Y with reliability ρ . Signal quality q (or equivalently δ) is the dial for Assumption A5.
- **Beliefs.** Because signals are conditionally independent given Y (A4), posterior log-odds are *additive*: a truthful Bayesian inverts the previous report into its log-odds and adds its own signal’s log-likelihood ratio ℓ . This is the computational heart of the simulator and is what makes exact aggregation possible.
- **Strategies.** A `Strategy` maps an agent’s context (prior, previous report, own signal, history) to a report. The truthful strategy reports the Bayesian posterior; the family we use to model deviation is `ScaledSignal( $\gamma$ )`, which reports $\text{logit}(\pi) + \gamma \ell$, so $\gamma = 1$ is truthful and $\gamma > 1$ is over-confident. Additional strategies (herding, noisy, constant, contrarian) are used in the population study.
- **Scoring rules.** `CrossEntropyMSR` implements (2); outcome-based benchmarks (`OutcomeLogMSR`) are available for validation and are used in the empirical study.
- **Market.** `SelfResolvingMarket` runs agents in sequence, terminates with probability α (or after a fixed count when $\alpha = 0$), designates the terminal agent as the reference, and applies the CE-MSR to the first $T - k$ agents and the flat fee R to the last k .

By construction the simulator satisfies A1 (agents are Bayesian; a deviator deviates *within* that rational frame via γ), A2 (one common prior generates every signal), and A3–A4 (distinct, conditionally independent signals). The only assumption we deliberately vary is A5 — how informative signals are, and how many of them the reference sees. This is what makes the simulator a clean test of the mechanism’s *core* incentive claim rather than a test of the modelling assumptions themselves.

4.2 RQ1: estimating the profit-maximising deviation

RQ1 asks whether a freely deviating agent prefers to report truthfully, and how that depends on the reference’s information. We isolate a single *focal* agent with signal quality $q_t = 0.7$ and let it consider the deviation family `ScaledSignal( $\gamma$ )` over a grid $\gamma \in [0, 3]$ (61 points). The reference observes the focal agent’s report *plus* k additional independent signals, each of quality $q_r = 0.65$; these are the reference’s informational substitutes. We sweep

$$k \in \{0, 2, 8, 32, 64, 128, 256\},$$

and for each (k, γ) estimate the focal agent’s expected CE-MSR payoff by Monte Carlo with $N = 4 \times 10^5$ samples (prior $\pi = 0.5$). For each k we report the profit-maximising deviation $\gamma^*(k) = \arg \max_{\gamma} \mathbb{E}[\text{payoff}]$.

This is the direct operationalisation of Theorem 3: the gain from deviating is bounded by $\hat{D}_{\eta}(\Delta, \mathbf{y})$, and $\Delta \rightarrow 0$ as k grows, so the profit-maximising γ^* should fall from “over-report as much as possible” toward the truthful value 1. The deviation parameter γ is simply how we *realise* a deviation; $\gamma^* \rightarrow 1$ is the theoretical bound biting.

4.3 Reproducing the analytic bound $k_{\min}$

To connect the Monte Carlo result to the paper’s closed-form theory, we also reproduce the analytic minimum-substitutes bound of Srinivasan et al. (2025) (their Figure 2): the smallest number of substitutes k the reference needs so that a focal agent’s gain from deviating under the CE-MSR

is at most ε . The computation chains three of the paper’s results: Theorem 1 (Eq. 3) bounds the adjustment $|\Delta|$ as a function of k ; Theorem 5 (Eq. 9) bounds the misreport gain $\hat{D}_\eta(\Delta, y)$ as a function of Δ and the market prior y ; and Remark 3 inverts these numerically — because $\hat{D}_\eta(\Delta, y) = \varepsilon$ is transcendental, we solve $\varepsilon' = \min\{|\Delta| : \hat{D}_\eta(\Delta, y) = \varepsilon\}$ by bisection (in a log-transformed variable, so roots sitting e^{-100} from the pole are still resolved) and substitute back into the Theorem 1 bound to obtain

$$k_{\min}(\delta, \eta, \varepsilon, y_1) = \frac{1}{-\log(1 - \delta)} \log\left(\frac{(1 - \eta)/\eta - \eta/(1 - \eta)}{4\varepsilon'}\right).$$

We evaluate $k_{\min}$ across a grid of market priors y_1 for representative $(\delta, \eta, \varepsilon)$, recovering the shape of the paper’s analytic curve and giving a theory-side reference point for the empirical onset of truthfulness seen in RQ1.

4.4 Supporting synthetic studies

Three further studies probe the mechanism’s aggregation behaviour rather than its incentives:

- **Aggregation.** Sweep the number of truthful agents and the signal quality q , and measure market accuracy, Brier score, and the *gap* between the terminal report and the ideal full-information Bayesian posterior. This checks the paper’s claim that, under truthful play, the market exactly aggregates.
- **Value of an informed reference.** Vary the fraction of near-oracle agents (reliability 0.99) in a 12-agent market, and separately the quality of a *single* informed agent, to see how much oracle access the aggregate needs.
- **Mixed / strategic populations.** Compare an all-truthful population against populations seeded with herders, noisy reporters, an uninformative third, and over-reporters, measuring accuracy, gap to full information, and the designer’s total cost. The cost in the uninformative population is the empirical check on the paper’s “pays zero in uninformative equilibria” claim.

4.5 RQ2: a counterfactual on real Polymarket data

RQ2 leaves the synthetic world entirely. We download roughly 90 *resolved* Polymarket markets (Polymarket 2025) with trading volume above \$100k via the Gamma and CLOB APIs (responses cached on disk for reproducibility). For each market we treat the Yes-price time series as a stream of sequential reports: a price p_t is the report $\mathbf{q}^{(t)} = [1 - p_t, p_t]$, and each price *change* is one “reporter” (flat repeats are dropped). We then pay every transition *two ways on the identical set of transitions*:

- **self-resolving:** CE-MSR (2) against a late *reference price* $\mathbf{r}$ — no outcome used;
- **verifiable:** log market scoring rule against the realised outcome Y .

Because the late reference price plays the role of the paper’s terminal agent, the choice of “how late” is the crucial knob, and it mixes two channels: a late price is both *better informed* and *contaminated by outcome leakage* (near resolution the question is almost settled). We therefore sweep several references: `lead_1h`, `lead_24h`, `lead_168h` (1h/1d/1w before the last data point), `life_50pct` (the price halfway through the market’s life), and `window_1-24h` (the variance-reducing *average* of a late window). To compare markets of very different durations on a common axis we additionally sweep the reference by *relative* time, from 90% before close (very early) to 1% before close (near resolution).

For each market and reference we compute: the Pearson and Spearman correlation of per-reporter payouts between the two schemes (do the *amounts* agree, and does the *ranking* of who earns most agree?), the Jaccard overlap of the top-decile earners, the ratio of total designer payout, and

whether the single most-rewarded reporter matches. We aggregate as medians across markets and also break results down by a reproducible keyword topic classifier (politics / sports / economy / tech / ...), since Polymarket’s own category fields are mostly empty.

What RQ2 does and does not measure. This design measures whether the self-resolving mechanism’s *outputs* (its payouts) track the verifiable mechanism’s outputs on real data. It is *not* a test of incentive compatibility: real traders are not the rational Bayesians of A1, prices are serially correlated rather than conditionally independent (A4 fails), and we replay a fixed historical path rather than letting agents deviate. RQ2 is therefore a robustness test — does reference-based resolution still reproduce ground-truth-based payouts when the assumptions behind it are broken? — and is complementary to the incentive question that RQ1 answers in the clean world.

5 Experiments and Results

We present RQ1 (incentives, Section 5.1), the supporting aggregation studies (Section 5.2), and RQ2 (robustness on real data, Section 5.3).

5.1 RQ1: truthfulness requires an informed reference

Figure 2 shows the focal agent’s expected-payoff landscape as a function of its deviation γ , one curve per number of reference substitutes k , together with the profit-maximising $\gamma^*(k)$. The pattern is exactly the one Theorems 1–3 predict:

- For $k \leq 8$ the best response is pegged at the grid maximum ($\gamma^* = 3$): with a poorly informed reference the focal agent profits by grossly over-reporting, because it can drag the reference toward its own report.
- At $k = 32$ the best response has fallen to $\gamma^* \approx 1.2$ — nearly truthful.
- For $k \geq 64$ the best response is *exactly* truthful, $\gamma^* = 1.0$: once the reference has enough informational substitutes, no deviation pays.

This is the paper’s central incentive claim, realised operationally: truthful reporting becomes a best response precisely when the reference is well-informed enough to act as a proxy for the ground truth. As a side check, under truthful play the terminal report reproduces the full-information Bayesian posterior to within a gap of $\sim 10^{-13}$, confirming the simulator implements the mechanism faithfully.

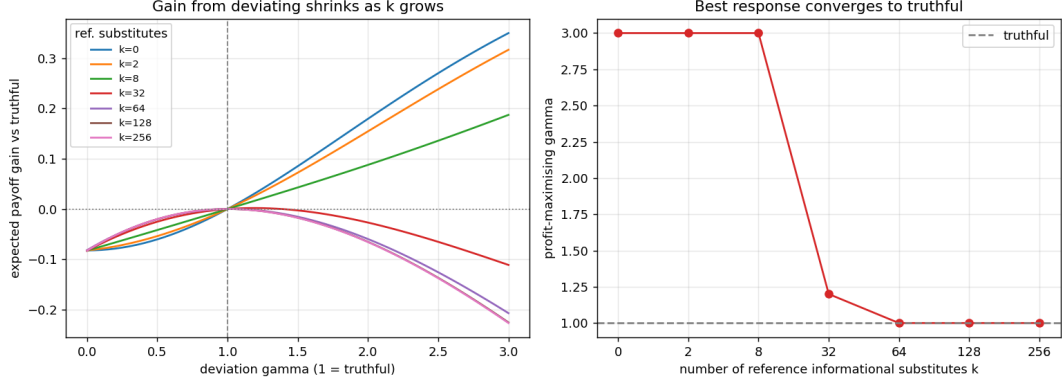


Figure 2: **RQ1.** Left: the focal agent’s expected-payoff gain over truthful reporting as a function of its deviation γ ; the gain from deviating shrinks as the reference’s number of informational substitutes k grows. Right: the profit-maximising γ^* converges to the truthful value 1 (dashed) by $k = 64$. Monte Carlo, 4×10^5 samples per point.

The analytic bound $k_{\min}$. Figure 3 reproduces the paper’s closed-form minimum-substitutes curve. For a stringent tolerance ($\delta = 0.1$, $\eta = 0.1$, $\varepsilon = 10^{-6}$), the bound sits at $k_{\min} \approx 146$ substitutes and is essentially flat across the market prior y_1 . Two things are worth noting. First, the analytic bound is considerably *looser* than the empirical onset in Figure 2: the Monte Carlo best response is already truthful at $k = 64$, whereas the worst-case bound demands ≈ 146 for a 10^{-6} -small gain — expected, since $k_{\min}$ is a worst-case guarantee over all priors and deviations at a far tighter ε . Second, both tell the same qualitative story — the number of substitutes needed grows mildly (like $1/(-\log(1 - \delta)) \cdot \log(1/\varepsilon)$) and is governed by signal informativeness δ .

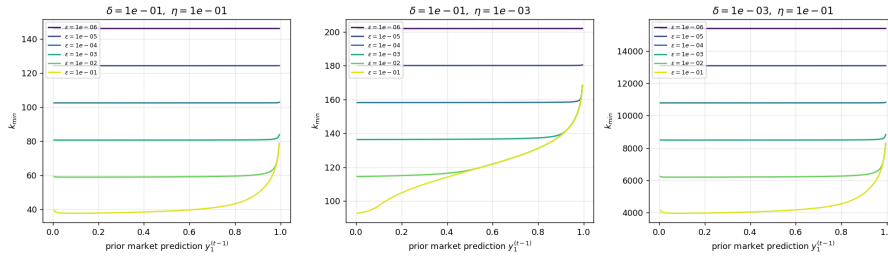


Figure 3: Analytic minimum-substitutes bound $k_{\min}$ as a function of the market prior y_1 , reproducing Figure 2 of Srinivasan et al. (2025) by chaining their Theorem 1, Theorem 5, and Remark 3. The bound is a worst-case guarantee and is looser than the empirical onset of truthfulness in Figure 2.

5.2 Supporting studies: aggregation behaviour

Exact aggregation under truthful play. With all agents truthful, the terminal report equals the full-information Bayesian posterior up to machine precision: the measured gap is $\sim 10^{-16}$ across signal qualities and market sizes. Accuracy rises with both the number of agents and signal quality — e.g. with quality $q = 0.8$ a 12-agent market reaches 98.7% accuracy, and with weak signals $q = 0.65$ a 30-agent market still reaches 95.4%. The mechanism aggregates exactly, as claimed.

A few near-oracles pin the aggregate. Table 1 shows that the reference does not need many oracles to be a good proxy for the truth. A *single* near-oracle agent (reliability 0.99) among 12 raises market accuracy from 56% to 99%, and once roughly five of twelve agents are near-oracles

the market is essentially perfect. This is the aggregation-side analogue of RQ1’s incentive result: “enough good information in the reference” is a mild requirement in practice.

Table 1: Market accuracy and Brier score as the number of near-oracle agents (reliability 0.99) in a 12-agent market grows. A single oracle already nearly resolves the market.

# near-oracles	oracle fraction	accuracy	Brier
0	0.00	0.561	0.244
1	0.08	0.990	0.0097
2	0.17	0.992	0.0046
3	0.25	0.999	0.0007
5	0.42	1.000	$\sim 10^{-8}$
12	1.00	1.000	$\sim 10^{-22}$

Mixed and strategic populations. Table 2 stress-tests aggregation when not everyone is truthful. Herders, noisy reporters, and uninformative agents all degrade accuracy and open a gap to full information, as expected. The most telling row is `uninformative_third`: when a third of agents report uninformatively, the designer’s total cost collapses to 0.046 — the mechanism *pays almost nothing in the uninformative regime*, the empirical counterpart of the paper’s claim that it pays zero in uninformative equilibria. Over-reporting, by contrast, is expensive for the designer (cost 0.506) while barely improving accuracy, which is the flip side of the RQ1 incentive to over-report when the reference is weak.

Table 2: Aggregate accuracy, gap to the full-information posterior, and designer cost across populations. The uninformative population pays almost nothing; over-reporting is costly without improving accuracy.

population	accuracy	Brier	gap to full-info	mean total cost
all truthful	0.845	0.105	$\sim 10^{-16}$	0.341
half herders	0.768	0.154	0.164	0.197
noisy	0.703	0.230	0.244	0.380
uninformative third	0.642	0.213	0.279	0.046
overconfident	0.852	0.112	0.074	0.506

5.3 RQ2: do self-resolving payouts track verifiable payouts on real data?

Reference sweep. Table 3 and Figure 4 report agreement between the self-resolving and verifiable payment schemes across 91 resolved Polymarket markets, as the reference is moved from near-resolution (`lead_1h`) to mid-life (`life_50pct`). At a genuinely late reference the two schemes are almost indistinguishable: at `lead_24h` the median Pearson correlation of per-reporter payouts is 1.00, the Spearman rank correlation is 0.997, the top-decile earners overlap perfectly, and the designer’s total payout is 95% of the verifiable total. As the reference moves earlier, the *amounts* keep tracking (Pearson stays ≥ 0.95) but the *ranking* of who earns most degrades sharply (Spearman falls to 0.58 at mid-life) and the total payout ratio drops to 0.56. That degradation *is* the outcome-leakage channel: a near-resolution reference rewards exactly the traders the outcome would, whereas a genuinely forecast-only reference cannot.

Table 3: Median agreement between self-resolving and verifiable payouts across 91 resolved Polymarket markets, by reference. Pearson asks whether payout *amounts* agree; Spearman whether the *ranking* of earners agrees; ratio is self-resolving total payout over verifiable total.

reference	Pearson	Spearman	top-decile Jaccard	payout ratio	argmax match
lead_1h	1.000	1.000	1.00	0.978	0.956
lead_24h	1.000	0.997	1.00	0.953	0.923
lead_168h	0.998	0.988	0.93	0.847	0.813
life_50pct	0.954	0.581	0.56	0.559	0.780
window_1-24h	1.000	0.997	1.00	0.957	0.923

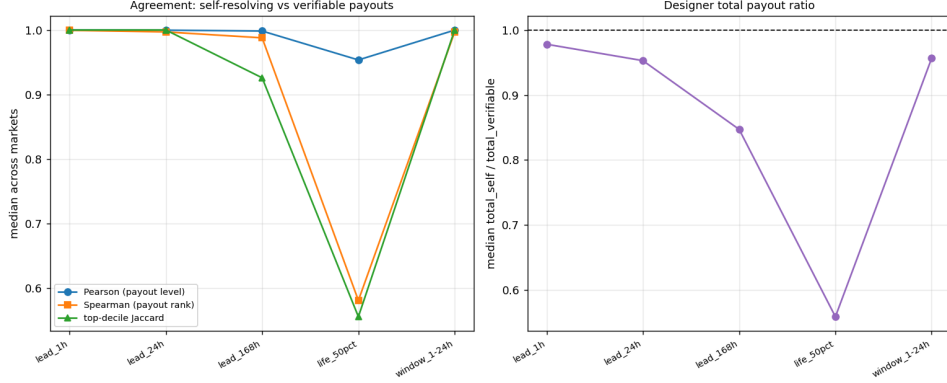


Figure 4: **RQ2, reference sweep.** Agreement between self-resolving and verifiable payouts as the reference moves from mid-life toward resolution. Amounts (Pearson) track throughout; ranking (Spearman) and the designer’s total cost (ratio) are what the outcome-leakage channel inflates near resolution.

Relative-time sweep. Because markets differ enormously in duration, Figure 5 sweeps the reference by *relative* time to close. Agreement rises monotonically as the reference approaches resolution: Pearson $0.94 \rightarrow 1.00$, Spearman $0.67 \rightarrow 1.00$, and payout ratio $0.48 \rightarrow 0.94$ as we move from 90%-before-close to 1%-before-close. The key honest takeaway is the left end of this curve: even at a genuinely *early* reference the payout *levels* still track well (Pearson ≈ 0.94); it is the *ranking* and the designer’s total cost that depend on how much the reference has already absorbed the outcome.

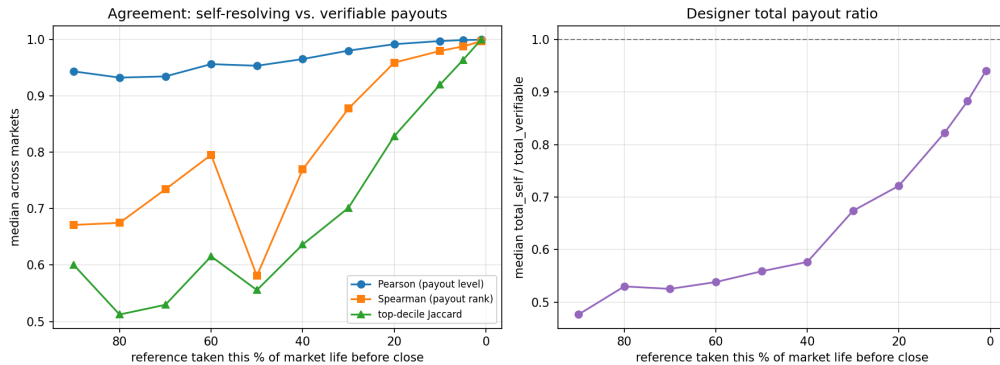


Figure 5: **RQ2, relative-time sweep.** Median agreement as the reference is placed from 90% before close (very early) to 1% before close (near resolution). All three measures improve as the reference approaches resolution.

Worked example and per-market detail. Figure 6 shows a representative high-volume market (the Kamala Harris popular-vote market, $\sim 2,700$ price moves, outcome $Y = 0$): the self-resolving payouts (CE-MSR against the 24h reference) and the verifiable payouts are visually indistinguishable (Pearson 1.00, Spearman 1.00, payout ratio 0.99). Table 4 makes the heterogeneity explicit. At a genuinely *late* reference every market tracks perfectly; at *mid-life*, some markets still track well (the Russia–Ukraine ceasefire and Poilievre markets), while others collapse and even invert (the US government shutdown and TikTok-ban markets, Spearman -0.48 and -0.54). The more a late price already leaks the outcome, the larger the mid-life gap — and topic breakdowns reflect this too (e.g. sports markets, which sit flat and then jump, keep high Pearson but low mid-life Spearman).

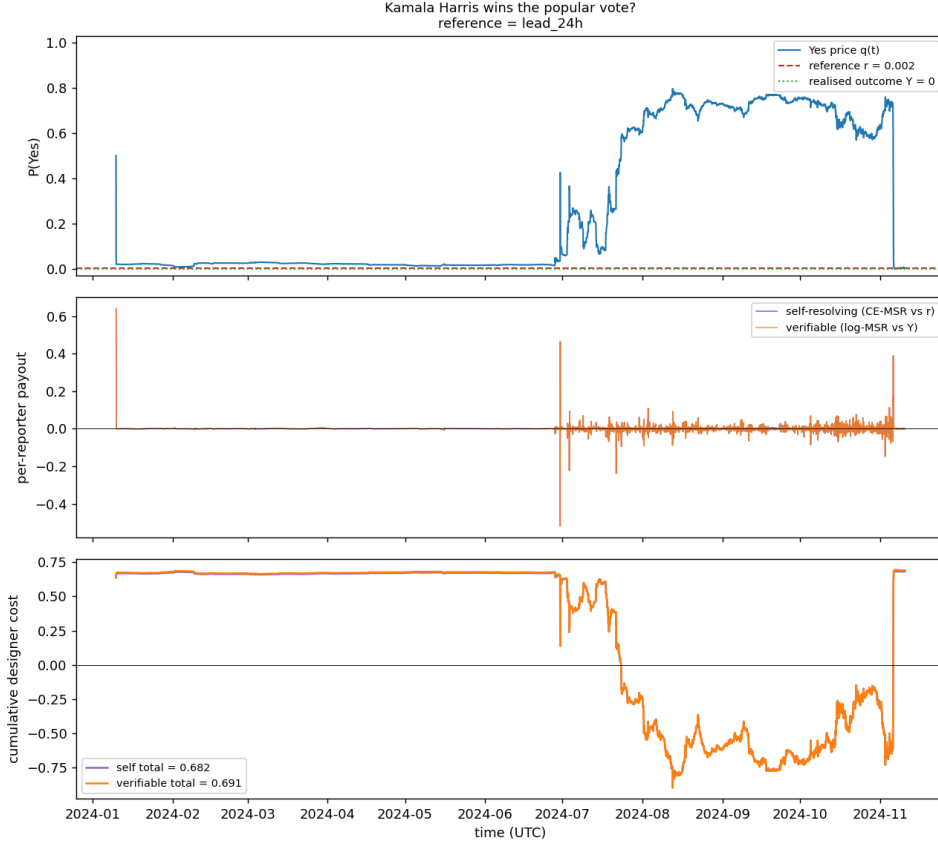


Figure 6: **RQ2, worked example.** Per-reporter payouts for the Kamala Harris popular-vote market under the self-resolving scheme (vs. a 24h reference) and the verifiable scheme (vs. the outcome). The two are nearly identical (Pearson 1.00, ratio 0.99).

Table 4: Pearson (P) and Spearman (S) correlation of per-reporter payouts for five high-volume markets, at a late (24h before close) and a mid-life (50%) reference. Late references track perfectly; mid-life references track only when the late price has not yet leaked the outcome.

market	# reporters	late (24h)		mid-life (50%)	
		P	S	P	S
Kamala popular vote	2681	1.00	1.00	1.00	0.45
Poilievre next Canadian PM	1030	1.00	1.00	0.96	0.91
Russia–Ukraine ceasefire '25	2531	1.00	1.00	0.99	0.98
US government shutdown	778	1.00	1.00	0.41	-0.48
TikTok ban (May '25)	1078	1.00	1.00	-0.37	-0.54

Summary of RQ2. On real data the self-resolving mechanism reproduces the verifiable mechanism’s payouts almost exactly when the reference is genuinely late, and degrades gracefully and interpretably as the reference moves earlier. The residual gap is concentrated in the *ranking* of earners and the designer’s *total cost*, and is driven by outcome leakage rather than by the mechanism failing to identify good forecasts — the payout *levels* track even at early references.

6 Conclusion, Discussion, and Limitations

6.1 Summary

This project gave empirical content to the self-resolving prediction market of Srinivasan et al. (2025), a mechanism that elicits and aggregates beliefs about an *unverifiable* outcome by paying agents the cross-entropy of their report against a late reference agent rather than against the truth. We reproduced the mechanism in an open-source agent-based simulator and evaluated it along two axes. In the clean synthetic world (RQ1), a freely deviating agent’s profit-maximising report converges onto truthful reporting as the reference accumulates informational substitutes ($\gamma^* \rightarrow 1$ by $k \approx 64$), operationalising the paper’s Theorems 1–3; we also reproduced the analytic minimum-substitutes bound $k_{\min}$ and confirmed exact aggregation under truthful play. On real Polymarket data (RQ2), the self-resolving payouts track the verifiable payouts they replace almost perfectly at a genuinely late reference (Pearson ≈ 1.00 , payout ratio ≈ 0.95), and degrade gracefully as the reference moves earlier, isolating outcome leakage as the channel driving the residual gap.

6.2 Discussion

The two studies answer two genuinely different questions, and it is worth being precise about which is which. RQ1 is about *incentives*: in a world where the paper’s assumptions hold, a rational agent who is free to lie chooses not to, once the reference is informed enough. RQ2 is about *outputs*: on real data where the assumptions fail, the payouts the mechanism produces still resemble the payouts an outcome-based mechanism would have produced. Neither subsumes the other — RQ2 is not an incentive test, and RQ1 is not a test on real behaviour — but together they bracket the mechanism from both sides and suggest it is more robust in practice than its worst-case theory promises (the empirical onset of truthfulness at $k \approx 64$ is well inside the $k_{\min} \approx 146$ worst-case bound).

The most interesting tension the empirical study surfaces is that the mechanism’s residual error lives almost entirely in the *ranking* of who deserves the most reward, not in the payout *levels*. A self-resolving designer can recover roughly the right total spend and reward roughly the right behaviour at an early, forecast-only reference; what they cannot fully recover without leaking the outcome is the fine-grained ordering of which trader moved the price toward truth. This points directly at a design question: can the mechanism be adjusted so that the ranking better matches who actually moved the price toward truth, without needing the outcome?

6.3 Limitations

Several limitations qualify these results and frame future work.

- **RQ2 measures outputs, not incentives.** We replay fixed historical price paths and never let agents deviate; the empirical study therefore says nothing about whether real traders would report truthfully under the live mechanism. A live or agent-in-the-loop deployment would be needed for that.
- **Broken assumptions on real data.** Real markets violate essentially all of A1–A5: traders are not risk-neutral Bayesians, there is no common prior, and price moves are serially cor-

related (herding, momentum, shared news) rather than conditionally independent given Y . The agreement we observe is thus evidence of *robustness*, not of the assumptions holding.

- **The reference conflates two things.** A late reference price is both better-informed and outcome-contaminated; our sweeps separate these channels only indirectly, by varying lateness. A cleaner instrument for “well-informed but not omniscient” would strengthen the claim.
- **Scope of the synthetic model.** The simulator deliberately makes A1–A4 hold by construction and only varies A5. This is the right design for isolating the core incentive claim, but it cannot speak to how the mechanism behaves when, say, conditional independence is mildly violated — a natural next experiment.
- **Binary outcomes only.** Both the mechanism as implemented and our studies are restricted to binary Y . Multi-outcome and continuous predictions are left for future work.

6.4 Future work

Two directions follow directly from the limitations. First, a *mechanism-design* question: can the CE-MSR be reweighted or the reference reconstructed so that the *ranking* of rewards tracks who pushed the price toward truth, closing the gap that RQ2 attributes to leakage — without observing the outcome? Second, a *generality* question: extending the simulator and the empirical replay to non-binary outcomes, and to information structures that depart from strict conditional independence, would test how far the reference-as-proxy idea stretches beyond the regime where it is provably truthful. More broadly, the gap between the loose worst-case $k_{\min}$ and the much earlier empirical onset of truthfulness suggests there is room for tighter, prior-dependent guarantees that better match observed behaviour.

Bibliography

- Aumann, Robert J. (1976). ‘Agreeing to Disagree’. In: *The Annals of Statistics* 4.6, pp. 1236–1239.
- Berg, Joyce E., Forrest D. Nelson and Thomas A. Rietz (2008). ‘Prediction Market Accuracy in the Long Run’. In: *International Journal of Forecasting* 24.2, pp. 285–300.
- Brier, Glenn W. (1950). ‘Verification of Forecasts Expressed in Terms of Probability’. In: *Monthly Weather Review* 78.1, pp. 1–3.
- Chen, Yiling, Stanko Dimitrov et al. (2010). ‘Gaming Prediction Markets: Equilibrium Strategies with a Market Maker’. In: *Algorithmica* 58.4, pp. 930–969.
- Chen, Yiling and David M. Pennock (2010). ‘Designing Markets for Prediction’. In: *AI Magazine* 31.4, pp. 42–52.
- Chen, Yiling and Bo Waggoner (2016). ‘Informational Substitutes’. In: *2016 IEEE 57th Annual Symposium on Foundations of Computer Science (FOCS)*, pp. 239–247.
- Gneiting, Tilmann and Adrian E. Raftery (2007). ‘Strictly Proper Scoring Rules, Prediction, and Estimation’. In: *Journal of the American Statistical Association* 102.477, pp. 359–378.
- Good, I. J. (1952). ‘Rational Decisions’. In: *Journal of the Royal Statistical Society: Series B* 14.1, pp. 107–114.
- Hanson, Robin (2003). ‘Combinatorial Information Market Design’. In: *Information Systems Frontiers* 5.1, pp. 107–119.
- Kong, Yuqing and Grant Schoenebeck (2019). ‘An Information Theoretic Framework for Designing Information Elicitation Mechanisms That Reward Truth-Telling’. In: *ACM Transactions on Economics and Computation* 7.1, pp. 1–33.
- (2022). ‘False Consensus, Information Theory, and Prediction Markets’. In: *13th Innovations in Theoretical Computer Science Conference (ITCS)*.
- Miller, Nolan, Paul Resnick and Richard Zeckhauser (2005). ‘Eliciting Informative Feedback: The Peer-Prediction Method’. In: *Management Science* 51.9, pp. 1359–1373.
- Polymarket (2025). *Polymarket: Decentralized Prediction Markets*. Polymarket. URL: <https://polymarket.com>.
- Prelec, Drazen (2004). ‘A Bayesian Truth Serum for Subjective Data’. In: *Science* 306.5695, pp. 462–466.
- Srinivasan, Siddarth, Ezra Karger and Yiling Chen (2025). *Self-Resolving Prediction Markets for Unverifiable Outcomes*. arXiv: 2306.04305 [cs.GT]. URL: <https://arxiv.org/abs/2306.04305>.
- Waggoner, Bo and Yiling Chen (2014). ‘Output Agreement Mechanisms and Common Knowledge’. In: *Proceedings of the Second AAAI Conference on Human Computation and Crowdsourcing (HCOMP)*.